CENTRAL PORT ELIZABETH URBAN

3D Spatial Twin & Volumetric Density Assessment · Donkin Corridor & NMU Campus

1. Executive Summary & Spatial Overview

This technical report presents the volumetric massing and spatial density profile for the Central Port Elizabeth (Gqeberha) precinct, focused on the Nelson Mandela University (NMU) Bird Street Campus and the adjacent Donkin Heritage Corridor (Bounding Box: 25.610° E to 25.625° E, -33.970° S to -33.960° S). Utilizing extracted 3D building vector footprints and height metadata, this analysis quantifies vertical enclosure, gross floor area (GFA), ground coverage, and volumetric mass density ratios to support urban planning, heritage preservation, and digital twin integration.

179,612 m³	22,270 m²	51,620 m²	1.60 m³/m²
Total Enclosed Volume	Footprint Area (2.23 ha)	Gross Floor Area (GFA)	Precinct Mass Density

2. Sub-Precinct Massing Breakdown

Nelson Mandela University (NMU) Bird Street Campus: The campus precinct comprises 2,805 m² of footprint area and 22,672 m³ of built volume across four core academic facilities. The Main Visual Arts building forms the anchor structure with 9,264 m³ (9.6m height, 3 storeys), supplemented by the Conservatoire & Music School (5,328 m³), Art Gallery & Exhibition Hall (4,960 m³), and 24-Hour Student ICT Center (3,120 m³).

Donkin Heritage Corridor: Dominated by historic spires and landmark monuments, this corridor exhibits the highest vertical mass density per square meter of footprint. Holy Trinity Anglican Church contributes 9,570 m³ across a 580 m² footprint (16.5m spire height), while the Donkin Reserve Lighthouse & Keeper House reaches a peak vertical height of 22.0m, yielding a vertical volumetric density of 22.0 m³/m².

Commercial & Mixed-Use Sector: High-density commercial student housing blocks along Bird Street (Annex A & Annex B) account for 32,230 m³ across 2,430 m² of ground area. Surrounding Victorian terrace rows and Central mixed-use residential blocks provide an additional 107,830 m³ of built massing, maintaining an average height profile of 6.8m to 7.0m.

3. Structure Massing & Density Metrics

Structure / Asset Name	Category	Footprint	Height	Volume	Density
Donkin Reserve Lighthouse	Heritage	185 m²	22.0 m	4,070 m³	22.0 m³/m²

Structure / Asset Name	Category	Footprint	Height	Volume	Density
Holy Trinity Anglican Church	Heritage	580 m²	16.5 m	9,570 m³	16.5 m³/m²
Bird St Housing Annex B	Commercial	1,280 m²	13.5 m	17,280 m³	13.5 m³/m²
Bird St Housing Annex A	Commercial	1,150 m²	13.0 m	14,950 m³	13.0 m³/m²
Donkin Stone Pyramid	Heritage	120 m²	10.0 m	1,200 m³	10.0 m³/m²
NMU Main Visual Arts	NMU Campus	965 m²	9.6 m	9,264 m³	9.6 m³/m²
NMU Art Gallery & Hall	NMU Campus	620 m²	8.0 m	4,960 m³	8.0 m³/m²
NMU Music School	NMU Campus	740 m²	7.2 m	5,328 m³	7.2 m³/m²
Belmont Terrace Rows	Residential	1,610 m²	7.0 m	11,270 m³	7.0 m³/m²
Central Mixed-Use Blocks	Residential	14,200 m²	6.8 m	96,560 m³	6.8 m³/m²
NMU Student ICT Center	NMU Campus	480 m²	6.5 m	3,120 m³	6.5 m³/m²
7 Castle Hill Museum	Heritage	340 m²	6.0 m	2,040 m³	6.0 m³/m²

4. Volumetric Mass Density Visualization

5. Methodology & Spatial Data Provenance

> **Data Pipeline Provenance:** Footprint geometries and 3D height attributes were extracted from OpenFreeMap vector tiles (layer: building, attributes: height, min_height, building:levels). Missing vertical attributes were derived using standard storey height multipliers (3.2m/level). All coordinates are projected in EPSG:4326 (WGS84). Volumetric metrics are computed via 2.5D polygon extrusion (Volume = Footprint Area × Effective Height).